

Research Interests

Diffusion language models, AI for mathematics, AI safety and interpretability, formal methods.

Research Experience

The University of Texas at Austin

Postdoctoral Fellow, Institute for Foundations of Machine Learning

2025 – Present

Advisors: Sanjay Shakkottai and Adam Klivans

Research: Diffusion language models; AI agents for mathematical research

SRI International

Research Intern

2022

Mentor: Susmit Jha

Research: Control and verification of deep equilibrium models

Nokia Bell Labs

Research Intern

2019

Mentor: Kedar Namjoshi

Research: Certified compilers for WebAssembly

Harvard University School of Engineering and Applied Sciences

Research Intern

2018

Mentors: Stephen Chong and Marco Gaboardi

Research: Program synthesis and formal semantics for R

Max Planck Institute for Software Systems

Research Intern

2017

Mentors: Rupak Majumdar and Damien Zufferey

Research: Formal verification of embedded systems

Education

University of Pennsylvania

Ph.D. in Computer and Information Science

2019 – 2025

Advisors: Rajeev Alur and Eric Wong

Thesis: Trustworthy Machine Learning: Specification, Verification, and Explanation

Yale University

B.S. in Computer Science and Mathematics

2015 – 2019

Advisor: Ruzica Piskac

Research: Program analysis and symbolic execution for Haskell

Publications

* denotes equal contribution.

Preprints and Technical Reports

- [1] Lyuxin David Zhang, Eric Wong, Surbhi Goel, and **Anton Xue**. LESSER: Post-Training Data Selection with Output-Layer Gradients. *arXiv preprint arXiv:2610.03702*, 2026.
- [2] **Anton Xue**, Litu Rout, Aditya Akella, Adam Klivans, Sujay Sanghavi, and Sanjay Shakkottai. dQwen3.5: Hybrid-Attention Diffusion Language Models. *arXiv preprint arXiv:2609.20751*, 2026.
- [3] Alan Li*, Rahul Saha*, **Anton Xue**, Swarat Chaudhuri, Adam Klivans, Pravesh Kothari, and Raghu Meka. Long-Horizon AI Research for Grothendieck Constant: A Case Study in Human-AI Mathematical Collaboration. *arXiv preprint arXiv:2608.11195*, 2026.

- [4] Rahul Saha*, Alan Li*, **Anton Xue**, Swarat Chaudhuri, Adam Klivans, Pravesh Kothari, and Raghu Meka. New Lower and Upper Bounds for the Grothendieck Constant. *arXiv preprint arXiv:2608.11158*, 2026.
- [5] Alan Li*, Rahul Saha*, **Anton Xue**, Swarat Chaudhuri, Adam Klivans, Pravesh Kothari, and Raghu Meka. The Grothendieck Constant is Less Than $\pi/(2 \log(1 + \sqrt{2})) - 10^{-5}$. *arXiv preprint arXiv:2606.03991*, 2026.
- [6] **Anton Xue***, Litu Rout*, Constantine Caramanis, and Sanjay Shakkottai. AnCoder: Anchored Code Generation via Discrete Diffusion Models. *arXiv preprint arXiv:2602.17688*, 2026.
- [7] Shreya Havaldar*, Weiqiu You*, Chaehyeon Kim, **Anton Xue**, Helen Jin, Marco Gatti, Bhuvnesh Jain, Helen Qu, Amin Madani, Daniel Hashimoto, Gary Weissman, Rajat Deo, Sameed Khatana, Lyle Ungar, and Eric Wong. T-FIX: Text-Based Explanations with Features Interpretable to eXperts. *arXiv preprint arXiv:2511.04070*, 2025.

Peer-Reviewed Publications

- [1] Rahul Saha*, Alan Li*, **Anton Xue**, Adam Klivans, Pravesh Kothari, Raghu Meka, and Swarat Chaudhuri. Optimizing Analytic Constants via AI-Guided Lean Proof Refinement. *Advances in Neural Information Processing Systems (NeurIPS)*, 2026.
- [2] Shailesh Sridhar, **Anton Xue**, and Eric Wong. Missingness Bias Calibration in Feature Attribution Explanations. *International Conference on Learning Representations (ICLR)*, 2026.
- [3] Andy Yang, Christopher Watson, **Anton Xue**, Satwik Bhattamishra, Jose Llarena, William Merrill, Emile Dos Santos Ferreira, Anej Svete, and David Chiang. The Transformer Cookbook. *Transactions on Machine Learning Research (TMLR)*, 2026.
- [4] Helen Jin*, **Anton Xue***, Weiqiu You, Surbhi Goel, and Eric Wong. Probabilistic Stability Guarantees for Feature Attributions. *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
- [5] Weiqiu You, **Anton Xue**, Shreya Havaldar, Delip Rao, Helen Jin, Chris Callison-Burch, and Eric Wong. Probabilistic Soundness Guarantees in LLM Reasoning Chains. *Empirical Methods in Natural Language Processing (EMNLP)*, 2025.
- [6] Helen Jin*, Shreya Havaldar*, Chaehyeon Kim*, **Anton Xue***, Weiqiu You*, Helen Qu, Marco Gatti, Daniel Hashimoto, Bhuvnesh Jain, Amin Madani, Masao Sako, Lyle Ungar, and Eric Wong. The FIX Benchmark: Extracting Features Interpretable to eXperts. *Journal of Data-centric Machine Learning Research (DMLR)*, 2025.
- [7] Thomas Zhang*, Behrad Moniri*, Ansh Nagwekar, Faraz Rahman, **Anton Xue**, Hamed Hassani, and Nikolai Matni. On The Concurrence of Layer-wise Preconditioning Methods and Provable Feature Learning. *International Conference on Machine Learning (ICML)*, 2025.
- [8] **Anton Xue***, Avishree Khare*, Rajeev Alur, Surbhi Goel, and Eric Wong. Logicbreaks: A Framework for Understanding Subversion of Rule-based Inference. *International Conference on Learning Representations (ICLR)*, 2025.
- [9] Xiayan Ji*, **Anton Xue***, Eric Wong, Oleg Sokolsky, and Insup Lee. AR-Pro: Counterfactual Explanations for Anomaly Repair with Formal Properties. *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- [10] Chang Zhu, Ziyang Li, **Anton Xue**, Ati Priya Bajaj, William Gibbs, Yibo Liu, Rajeev Alur, Tiffany Bao, Hanjun Dai, Adam Doupe, Mayur Naik, Yan Shoshitaishvili, Ruoyu Wang, and Aravind Machiry. TYGR: Type Inference on Stripped Binaries using Graph Neural Networks. *USENIX Security*, 2024.
- [11] **Anton Xue**, Lars Lindemann, and Rajeev Alur. Chordal sparsity for SDP-based neural network verification. *Automatica*, 2024.
- [12] **Anton Xue**, Rajeev Alur, and Eric Wong. Stability Guarantees for Feature Attributions with Multiplicative Smoothing. *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [13] **Anton Xue**, Lars Lindemann, Alexander Robey, Hamed Hassani, George Pappas, and Rajeev Alur. Chordal Sparsity for Lipschitz Constant Estimation of Deep Neural Networks. *Conference on Decision and Control (CDC)*, 2022.
- [14] Osbert Bastani, Shuo Li, and **Anton Xue**. Safe Reinforcement Learning via Statistical Model Predictive Shielding. *Robotics: Science and Systems (RSS)*, 2021.

- [15] Rajeev Alur, Phillip Hilliard, Zachary Ives, Konstantinos Kallas, Konstantinos Mamouras, Filip Niksic, Caleb Stanford, Val Tannen, and **Anton Xue**. Synchronization Schemas. *Symposium on Principles of Database Systems (PODS)*, 2021.
- [16] **Anton Xue** and Nikolai Matni. Data-Driven System Level Synthesis. *Learning for Dynamics and Control (L4DC)*, 2021.
- [17] Kedar Namjoshi and **Anton Xue**. A Self-certifying Compilation Framework for WebAssembly. *Verification, Model Checking, and Abstract Interpretation (VMCAI)*, 2021.
- [18] William Hallahan, **Anton Xue**, and Ruzica Piskac. G2Q: Haskell constraint solving. *International Symposium on Haskell*, 2019.
- [19] William Hallahan, **Anton Xue**, Maxwell Troy Bland, Ranjit Jhala, and Ruzica Piskac. Lazy counterfactual symbolic execution. *Programming Language Design and Implementation (PLDI)*, 2019.

Mentoring Experience

Quentin Fruytier (UT Ph.D. student): Faster inference with diffusion language models	2026 – Present
Alan Li (UT Ph.D. student): AI agents for mathematical research	2025 – Present
Rahul Saha (UT Ph.D. student): AI agents for mathematical research	2025 – Present
David Zhang (Penn undergraduate): Post-training data selection for language models	2023 – Present
Faraz Rahman (Penn undergraduate): Explainability and interpretability of diffusion models	2023 – 2024
Jawad Ahmad (high school student): Programmer-friendly code synthesis with ChatGPT	2023

Awards & Fellowships

Postdoctoral Fellowship, Institute for Foundations of Machine Learning at UT Austin	2025
Recipient, AWS gift supporting trustworthy AI research at Penn (ASSET Center)	2023
University of Pennsylvania ENIAC Fellowship	2019
Yale Computer Science Award	2019
National Science Foundation Graduate Research Fellowship	2019
Yale College Freshman Summer Research Fellowship	2016

Professional Service

Area Chair: ICLR	2026 – Present
Reviewer: NeurIPS, ICML, ICLR, AAAI	2024 – Present
Reviewer: AI-MATH, AI-Native Academia, SATA, SciForDL, AdvML	2023 – Present
Reviewer: IEEE L-CSS, CDC, ACC, Automatica, L4DC	2021 – Present
Artifact Evaluation Committees: TACAS (2023), SAS (2021), VMCAI (2021), PLDI (2020, 2021)	2020 – 2023

Teaching Experience

University of Pennsylvania , Teaching Assistant: Convex Optimization; Fundamentals of Linear Algebra and Optimization; Mathematical Foundations of Computer Science	2020 – 2023
Yale University , Teaching Assistant / Course Grader: Real Analysis; Intensive Algorithms; Design and Analysis of Algorithms; Mathematical Tools for Computer Science	2016 – 2019